

PRE-REGISTERED TEST REPORT · PUBLIC

Layer 3 Onset Test and First Five-Layer Layup Sample

Bench Test 2 of the BTSB pre-Gen-0 programme

Report number	CIP-BTSB-BT-002
Revision	A — pre-registration (contains no results)
Issued	22 September 2026
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Repository	github.com/Idreesh111/btsb-bench-test-preregistration — folder BT-002

Document status

This document was completed and dated before any Bench Test 2 material was received and before any data were collected.

Sections 1–8 fix the apparatus, method, quantitative predictions, analysis and falsification criteria. Section 9 is blank at issue and is completed after testing. The purpose is the same as for BT-001: to fix the predictions and the analysis before the results are known, so that neither can be adjusted afterwards to fit the outcome.

Revision	Date	Status	Contains results?
A	22 Sep 2026	Pre-registration	No
B	___ Oct 2026	Final	Yes

[PRE-GEN-0 MECHANISM DEMONSTRATION AND LAYUP SAMPLE — NOT A PRODUCT VALIDATION]

1. Purpose

Bench Test 2 has four parts:

Part	Question
A	At what temperature does the expandable graphite supplied as BTSB Layer 3 begin to expand, and by how much does it expand freely? This completes the test pre-registered in BT-001 Rev A §8 (hypothesis H4) and deferred because the material had not been obtained.
B	Can the five-layer architecture be assembled by hand at the production footprint of 173 × 71 mm, and what thickness and mass does a hand-laid sample reach against the design values of 1.94 mm and 14.08 g?
C	When a hand-laid five-layer sample is placed in the BT-001 fixture, does the sample containing paraffin show a latent-heat plateau that an otherwise identical sample without paraffin does not?
D	Is there any conductive path through the finished samples, at multimeter test voltage, before and after heating?

2. What this test does NOT do

Stated first, so that nothing later can be read as claiming more than it does.

- **It does not test BTSB as a product.** The samples are hand-laid, not laminated, and use substitute materials (table below).

- **No cells. No thermal runaway. No flame test** (UL 94 is not performed). **No gas-adsorption measurement. No dielectric-strength measurement** — Part D is a low-voltage continuity check only.
- **Part A measures free expansion** of loose flakes. Expansion inside a clamped 0.40 mm product layer is lower and is not measured, except indicatively in optional Part A2.
- **Part C stays below 120 °C.** Layer 3 does not activate in Part C. Part C tests only how the assembled stack absorbs heat.
- **Layers 1, 4 and 5 are built but not functionally tested.** Layer 4's zeolite is used as received, not dehydrated, and is not expected to be functional for gas capture.
- **Not accredited, not independently witnessed, single operator.**

Materials used, against the design

Layer	Function	Design	Used in BT-002	Status
L1	Spreads heat sideways	Pyrolytic graphite, 0.05 mm	Commercial thermal graphite sheet	Substitute
L2	Absorbs heat (phase change)	RT64HC in expanded graphite, 1.00 mm	Paraffin wax in expanded graphite made from the Part A material	Substitute PCM
L3	Blocks flame and heat	Expandable graphite, 0.40 mm	Expandable graphite (grade as supplied)	Same material class
L4	Captures toxic gas	Zeolite 13X + activated carbon, 0.30 mm	Zeolite 13X + activated carbon, as received	Same class, not conditioned
L5	Electrical isolation	PVDF film, 0.15 mm	Polyimide (Kapton) film, adhesive-backed	Substitute

Design values per build manual and ECN-001. Binders, forming and joining methods differ from the design and are recorded in the confidential annex (Section 12).

3. Relation to BT-001

3.1 A deferred test, now performed

BT-001 Rev A §8 pre-registered an intumescent onset test and hypothesis H4, and recorded that the test was deferred because the material had not been obtained. It stated: *"It will not be quietly removed."* **Part A of BT-002 performs that test. H4 and its falsification criteria are carried over verbatim and unchanged** (Section 8).

The phase-change band against which H4 is judged is the one BT-001 measured: the Run A plateau, with TC1 rising from **60.5 °C to 69.3 °C** (BT-001 final report §9.1).

3.2 Three method changes carried forward

BT-001's final report (§14) named three changes for the next stage. All three are applied here:

BT-001 lesson	How BT-002 applies it
Melt the PCM in place before the run — cast wax pulled away from the plates and added contact resistance	Each sample receives one conditioning run in the clamped fixture before measurement. The paraffin is also held inside an expanded-graphite matrix rather than cast solid.
Solve containment before the programme — the foil liner created a 44 % thermal bypass	No gasket and no foil liner. The paraffin is held by capillary action in the matrix, and the sample edges are sealed with polyimide tape.
n = 3 minimum per condition	Three measured runs of each sample in Part C; three runs in Part A.

3.3 Analysis methods fixed in advance this time

BT-001's principal results came from a direct climb-rate comparison against the control (final report §9.1) and an energy balance using a heat model fitted to the control (final report §9.2). BT-002 pre-registers **those same methods**, adapted only where stated in Section 6, so that this time they are fixed before the data exist.

4. Apparatus

4.1 Part A — onset

Item	Specification
Heat source	Electric hotplate, outdoors, slow ramp
Vessel	Steel, straight vertical walls. Not glass.
TC1	K-type, held against the outside of the vessel near its base with polyimide tape — reads hotter than the sample (upper bound)
TC2	K-type, buried in the middle of the flake bed — the measurement
Logger	BT-001 Arduino Uno + 2 × MAX6675, 1 Hz CSV, same sketch, not re-uploaded. Its 120 °C advisory alarm on TC1 will print in Part A and is ignored.
Volume	Marked 10 mL measure (V1); marked 500–1000 mL container (V2)
Record	Phone video with the vessel and the laptop screen in frame; operator declares observations aloud

4.2 Part B — samples

0.01 g digital balance (as BT-001) · digital vernier caliper · two spare aluminium plates from the BT-001 batch (173 × 71 × 3 mm), used as template and as thickness platens. Construction per confidential annex.

4.3 Part C — the BT-001 fixture

Unchanged from BT-001 except as listed: plates H 98.24 g and C 97.83 g (5052-H32, 173 × 71 × 3 mm; C_plates = 172.5 J/K); three 12 V cartridge heaters in series on plate H; 12 V supply connected directly, rocker switch abort; four G-clamps at 43 / 60 / 105 / 130 mm, finger tight + ½ turn; ceramic fibre insulation as rebuilt in BT-001 §7.1; heated from above; TC1 on plate H outer face, TC2 on plate C outer face; supply voltage and current recorded at the start and end of every run.

Changed from BT-001	Reason
No silicone gasket, no foil liner	The sample occupies the full plate area; the foil bypass is removed
Thin thermal-paste film on both sample faces	Same contact method as BT-001 Run C; the edge tape would otherwise hold the plates slightly off the sample
Sample orientation: L1 towards plate H (heated side)	Matches the design order, L1 facing the heat source

4.4 Part D — continuity

Digital multimeter on its highest resistance range (the range is recorded). The sample is placed between two clean spare aluminium plates acting as electrodes, under about 1 kg.

5. Method

5.1 Material verification (before Part A)

Every material is photographed with its label on receipt, with its invoice and any datasheet kept. The graphite is verified as **expandable**, **not expanded**, by a filmed 30-second check: a pinch on a hot steel surface must visibly expand. Supplier-stated onset and expansion figures are recorded where available; where none is supplied, this is stated.

5.2 Part A — onset

- 5 mL of loose flakes, tapped three times (V1), levelled in the steel vessel. TC1 on the vessel base, TC2 buried mid-bed.
- Slow ramp on the hotplate. The operator declares **first visible movement** aloud on video, with the TC2 reading, then rapid expansion, then completion.

3. Hotplate off at completion. Worms poured loose, uncompacted, into the marked container (V2).
4. **n = 3.** A run is valid if TC1 – TC2 at onset is $\leq 40\text{ }^{\circ}\text{C}$. A run exceeding this is reported, marked invalid for the onset figure, and repeated at a slower ramp.

5.3 Part A2 — constrained expansion (optional, indicative)

About 2 g of flakes spread between two spare aluminium plates under a weighed brick; stack height measured at four corners before and after heating past onset. Constrained ratio = height increase $\div$ initial layer thickness. Reported as indicative only.

5.4 Part B — samples

Sample	Composition
S1	All five layers, L1 $\rightarrow$ L5. L2 contains paraffin, target 5.5 g , measured by weighing before and after loading (m_w).
S2	Identical to S1 in every respect except that L2 contains no paraffin. Same material batches, same days, same handling — including the warming and pressing steps.

Mass to 0.01 g. **Thickness:** the sample is placed between the two spare plates; the stack is measured by vernier at four corners (15 mm from each corner); the plates' own thicknesses at the same points are subtracted; the result is the **mean of the four corners**. Mass is recorded after each layer. Finished samples are photographed on both faces beside a ruler and in edge view.

5.5 Part C — runs

```
S2 conditioning (logged, published, not analysed)
S2-1  S2-2  S2-3
-- sample change --
S1 conditioning (logged, published, not analysed)
S1-1  S1-2  S1-3
```

- **Conditioning:** heat until TC1 = $80\text{ }^{\circ}\text{C}$, power off, cool under clamp to ambient. Melts the paraffin in place in S1; applied identically to S2 so the two samples receive the same history.
- **Order:** the paraffin-containing sample runs last, as in BT-001.
- **Start condition:** both channels within $\pm 2\text{ }^{\circ}\text{C}$ of ambient; at least 60 s of baseline logged before power-on; per-run TC1 – TC2 baseline recorded. No fixed offset correction.
- **Termination** — first of: TC2 = $85\text{ }^{\circ}\text{C}$; TC1 = $110\text{ }^{\circ}\text{C}$ (early stop); TC1 = $120\text{ }^{\circ}\text{C}$ (hard abort); 45 min; any safety abort. Logging continues 10 min after power-off.
- Sample thickness is re-measured after Part C.

5.6 Part D — continuity

D0: the two electrode plates face to face must read near $0\text{ }\Omega$ (setup check). **D1:** S1 and S2 each measured between the plates before Part C. **D2:** repeated after Part C. Readings of partial stacks during construction are recorded as found, without prediction.

6. Pre-specified analysis

Fixed before data collection. No alternative analysis will be substituted after seeing the data.

6.1 Part A

```
T_on   = TC2 at declared first visible movement, per valid run
Result = mean of valid runs, with the range (min - max), n stated
Lag    = TC1 - TC2 at onset, per run
R_free = V2 / V1 per run; mean reported, with +/-20 % method uncertainty
```

6.2 Part B

Thickness t (mean of four corners) and mass m for S1 and S2; paraffin loading m_w ; each compared with the design (1.94 mm, 14.08 g) as a ratio.

6.3 Part C — climb-rate comparison

Adapted from BT-001 final report §9.1. **One-minute windows are used instead of two**, because the expected plateau is shorter.

$r(t)$ = TC1 climb rate in consecutive 1-minute windows from power-on
 r_{ref} = mean of the S2 runs, window by window

PLATEAU = 2 or more consecutive windows in which
 $r_{S1} \leq 0.50 \times r_{ref}$ AND $50\text{ C} \leq TC1(S1) \leq 78\text{ C}$

Duration = number of windows $\times$ 60 s

Band = TC1 at the start and the end of the plateau

Control check: the same detector is applied to each S2 run, using the mean of the *other two* S2 runs as its reference. A plateau "detected" in S2 counts against H6.

6.4 Part C — energy balance

Adapted from BT-001 final report §9.2. A heat model is fitted to each S2 run and the mean P and k are used:

Fit (S2): $C_{S2} \times dT_{mean}/dt = P - k \times (T_{mean} - T_{amb})$
 least squares, over $40\text{ C} \leq T_{mean} \leq 78\text{ C}$

$C_{S2} = 172.5\text{ J/K} + m(S2) \times 0.8\text{ J/g.K}$ (sample, approximate)
 $C_{S1} = C_{S2} + m_w \times 2.1\text{ J/g.K}$ (paraffin, sensible)

For each S1 run:

$E_{unacc}(t) = \text{integral}[P - k(T_{mean} - T_{amb})] dt - C_{S1} \times dT_{mean}$
 E_{latent} = value at which E_{unacc} stops accumulating
 (read as the mean of E_{unacc} over the 5 min after the plateau ends)

Agreement = $E_{latent} / (m_w \times 200\text{ J/g})$
 also reported at 180 and 220 J/g

6.5 Repeats

If two repeats of the same condition differ by more than **15 %** in plateau duration or in E_{latent} , they are **reported separately and not averaged**, as in BT-001.

6.6 Part D

Each reading recorded as "OL" (over range) or as a value, with the range used.

7. Predictions

Stated before data collection.

#	Quantity	Predicted
P1	Part A onset T_{on} (TC2, mean of valid runs)	150 – 200 °C (carried from BT-001 §7.2)
P2	Free expansion ratio R_{free} , loose poured volume	30 – 150 ×
P3	Constrained ratio (A2, if performed)	Lower than R_{free} of the same material
P4	S1 thickness, mean of four corners	2.0 – 5.0 mm
P5	S2 thickness, mean of four corners	2.0 – 5.0 mm
P6	S1 mass	14 – 28 g

#	Quantity	Predicted
P7	S2 mass	S1 mass minus m _w , within ± 3 g
P8	Paraffin loading m _w in S1	5.0 – 6.0 g
P9	S1 plateau detected (§6.3)	In at least 2 of 3 runs
P10	S1 plateau duration	2 – 6 min (120 – 360 s)
P11	S1 plateau band (TC1)	Within 55 – 72 °C
P12	S2 plateau detected (control)	In no more than 1 of 3 runs
P13	Energy balance agreement (§6.4), mean of S1 runs	50 – 150 %
P14	D1 and D2, S1 and S2	OL on the highest range in every case
P15	Thickness change after Part C	Decrease of 0 – 30 %

7.1 Derivations

P4 — why the design thickness is not expected

L1 graphite sheet	0.03 – 0.10 mm	
L2 hand-pressed matrix (+ paraffin)	1.0 – 2.0 mm	
L3 one layer of flakes in binder	0.3 – 0.8 mm	
L4 one layer of powder	0.1 – 0.4 mm	
L5 polyimide film with adhesive	0.05 – 0.10 mm	
bond lines, edge tape, unevenness	0.2 – 0.5 mm	
-----	-----	
total	~1.7 – 3.9 mm	-> predicted 2.0 – 5.0 mm

The **1.94 mm design thickness depends on industrial lamination. A hand layup is predicted, in advance, not to reach it.** A result above 1.94 mm is therefore not evidence against the design thickness; a result inside 2.0–5.0 mm confirms the prediction.

P10 — plateau duration

m _w = 5.5 g	x 200 J/g	= 1,100 J
x 0.85 (BT-001 energy-balance closure)		~ 935 J
BT-001 absorbed ~3,900 J over a 14-min plateau		~ 4.6 W mean absorption rate
Allowing 3 – 8 W for the thinner layer:		
935 J / 8 W	~ 120 s	935 J / 3 W ~ 310 s
With m _w from 5.0 to 6.0 g		-> predicted 2 – 6 min

P6 — mass

L1 ≈ 0.5–2.5 g; L2 matrix ≈ 2.5–4 g plus 5.5 g paraffin; L3 flakes and binder ≈ 2–5 g; L4 powder ≈ 1–4 g; L5 ≈ 1–2 g; edge tape ≈ 0.5 g. Total ≈ 13–23 g, widened to **14–28 g** for hand-layup variability. The design is 14.08 g.

8. Hypotheses and falsification criteria

What result would show each hypothesis to be wrong. Stated before data collection.

#	Hypothesis	Falsified if
H4 (from BT-001, verbatim)	The intumescent activates at a temperature that does not overlap the phase-change band	Onset below 100 °C , or the two measured bands overlap
H5	The supplied material is expandable graphite and expands freely by at least an order of magnitude	No expansion in the 30-second check, or mean R _{free} < 10 ×
H6	Within an assembled five-layer layup, the paraffin-containing sample shows a latent-heat plateau that the no-paraffin control does not	S1 plateau detected in fewer than 2 of 3 runs; or S2 plateau detected in 2 or more of 3 runs; or mean energy agreement outside 50 – 150 %

#	Hypothesis	Falsified if
H7	The finished layup has no conductive path through its thickness at multimeter test voltage	Any finite reading on the highest resistance range, for S1 or S2, before or after Part C

For H4, the phase-change band is BT-001's measured 60.5–69.3 °C; the intumescent band is the range of valid onset values from Part A.

8.1 Results that would NOT falsify

- A sample thicker than 1.94 mm — predicted in advance (P4, §7.1).
- A plateau shorter or longer than P10, with the energy balance inside 50–150 %.
- An onset that disagrees with the supplier's datasheet — a finding about the material.
- A Part A run invalidated by thermal lag and repeated — reported, not hidden.

8.2 Commitment

If any falsification criterion is met, that outcome will be reported in Revision B and in all derived material, with no change to the criteria above. Every prediction in Section 7 will be scored, including those that fail.

9. Results

To be completed after testing. Blank at pre-registration.

9.1 Part A

	A1	A2	A3	Result
Onset T _{on} (TC2, °C)				
TC1 – TC2 at onset (°C)				
Valid?				
R _{free} (×)				
Constrained ratio, A2 (×)				

9.2 Part B

	S1	S2	Design
Thickness, mean of 4 corners (mm)			1.94
Mass (g)			14.08
Paraffin m _w (g)		—	—
Thickness after Part C (mm)			—

9.3 Part C

Run	Plateau detected?	Duration (s)	Band TC1 (°C)	E _{latent} (J)	Agreement (%)
S2-1				—	—
S2-2				—	—
S2-3				—	—
S1-1					

Run	Plateau detected?	Duration (s)	Band TC1 (°C)	E_latent (J)	Agreement (%)
S1-2					
S1-3					

9.4 Part D

	D1 (before C)	D2 (after C)	Range
S1			
S2			

9.5 Aborted, failed or excluded runs

All runs attempted are listed here, including conditioning runs and any excluded run, with the reason.

10. Limitations

Applicable regardless of outcome.

1. Substitute materials in L1, L2 and L5; L4 not conditioned. Hand layup, not lamination.
2. No cells, no thermal runaway, no flame, gas or dielectric-strength test.
3. Part A measures free expansion by loose poured volume ($\pm 20\%$). Part A2, if performed, is a crude constrained measurement under a small load.
4. Onset is judged visually and declared aloud; thermal lag between vessel and sample is quantified but not eliminated.
5. Part C inherits the BT-001 fixture's heat loss and uncalibrated power measurement. The sample's sensible heat capacity is approximated ($0.8 \text{ J/g}\cdot\text{K}$).
6. Paraffin latent heat is assumed at 200 J/g (literature range $180\text{--}220$); no DSC measurement.
7. $n = 3$ per condition — better than BT-001, still small.
8. Absolute temperatures carry the $\pm 2.5^\circ\text{C}$ K-type Class 2 tolerance.
9. Part D uses multimeter voltage only (typically below 10 V); it says nothing about behaviour at pack voltages.
10. Single operator; not accredited; not independently witnessed.

11. Claim ceiling

What this test can support, depending on the results

- "The expandable graphite supplied as Layer 3 began to expand at $X \pm Y^\circ\text{C}$ ($n = 3$), measured by a thermocouple embedded in the flake bed. Free expansion of loose flakes was $Z\times$. Expansion in the constrained product geometry is lower."
- "Two BTSB mechanisms have now been measured at non-overlapping temperature bands: phase change at $60.5\text{--}69.3^\circ\text{C}$ (BT-001) and intumescent onset at $X^\circ\text{C}$ (BT-002)." — only if H4 is not falsified.
- "A hand-laid five-layer layup sample was built at the production footprint, with declared substitute materials. It measured $T \text{ mm}$ and $M \text{ g}$, against a laminated design of 1.94 mm and 14.08 g ."
- "In the BT-001 fixture, the layup containing paraffin showed a plateau of $D \text{ s}$ at $X\text{--}Y^\circ\text{C}$, which the identical sample without paraffin did not." — only if H6 is not falsified.
- "No conductive path was found through the finished samples at multimeter voltage, before or after heating." — only if H7 is not falsified.

What this test does not support — in any outcome

- "BTSB prototype tested" · "flame propagation blocked" · "gas capture demonstrated" · "dielectric performance validated"

- "1.94 mm achieved" (unless actually measured) · "product validated" · any free-expansion figure presented as the product's 10–20× figure

Every figure derived from this test carries: **[PRE-GEN-0 MECHANISM DEMONSTRATION AND LAYUP SAMPLE — NOT A PRODUCT VALIDATION]**. All build-manual performance figures remain **[GEN-0 VALIDATION REQUIRED]**.

12. Confidential annex and document fingerprints

The construction method for the Part B samples, and the operator's working guide, are **withheld from publication** until the complete specification of Indian patent application **IN 202641099975** (filed 19 August 2026; complete specification due 19 August 2027) has been filed. They contain construction detail that is not yet protected.

So that their contents and date can still be verified, their **SHA-256 fingerprints** are published here. When the documents are released, anyone can compute the fingerprint of the released file and confirm it matches — proving the documents existed, unchanged, on the date of this commit.

Withheld document	SHA-256 of the PDF file
BT-002_Doc1A_Private_Build_Method.pdf (Annex A — construction method)	f070c2f2ed1a2ab32883012cab1a2c628729f6e1a413fc494ef459f0757baf53
BT-002_Doc2_Working_Guide_PRIVATE.pdf (operator working guide)	ae51e716db6f6be01f0659e33669c64d759a8e9f015808143cb9af0a6cbf1498

The predictions, analysis and falsification criteria — everything needed to judge the result — are all in this public document. Nothing in the withheld documents changes them.

13. Data availability

Published after testing, in folder **BT-002** of github.com/Idreesh111/btsb-bench-test-preregistration: all raw CSV files unedited (including conditioning, aborted and invalid runs), Part A videos, completed record sheets, and photographs of the finished samples. Photographs of intermediate construction steps are withheld with the annex (Section 12).

14. Signature

Prepared by	S. Mohammed Idreesh
Position	Founder, CIP — Compute Infrastructure Platform
Signature	
Date of pre-registration (Rev A)	22 September 2026
Second person present (Part A)	
Date of final issue (Rev B)	

Report CIP-BTSB-BT-002 Rev A · Pre-registration · Contains no results. Governing values per ECN-001. All build-manual performance figures remain **[GEN-0 VALIDATION REQUIRED]**.